

STRUCTURED LITERATURE REVIEW

From Local Nanoplasmonics to Mesoscopic Optical Response

Classical models, their nanoscale breakdown, many-body benchmarks, Feibelman surface response, and quantum hydrodynamics

Critical review and model hierarchy through the 2026 anchor paper

CENTRAL QUESTION

How far can a nanostructure be scaled before a bulk local permittivity, a sharp interface, and classical Maxwell boundary conditions cease to be predictive—and which level of quantum or mesoscopic theory should replace them?

The review is organized as a controlled loss of approximation. It begins with the classical local-response platform for metals, films, graphene, and excitonic 2D layers; follows the experimental failure of that platform under strong confinement; then compares fully microscopic many-body methods with the two principal mesoscopic routes: Feibelman surface-response functions and quantum hydrodynamics. The final section shows exactly where the 2026 volume-integral SC-HDM paper enters this chain.

Level	Response object	Retained physics
1. Local classical	$\epsilon(\omega)$, $\sigma_{2D}(\omega)$	Bulk dispersion, geometry, radiation, absorption
2. Microscopic	$\chi(r, r'; \omega)$, many-electron states	Atomic structure, transitions, tunnelling, correlations
3. Surface mesoscopic	$d_{\perp}(\omega)$, $d_{\parallel}(\omega)$	Charge/current centroids and surface-enabled loss
4. Bulk nonlocal	$\epsilon(k, \omega)$ or hydrodynamic fields	Finite compressibility and longitudinal response
5. Self-consistent continuum	$n_0(r)$, $G[n]$, induced n_1/J	Diffuse equilibrium surface plus nonlocal dynamics

Executive summary

Local classical electrodynamics remains the correct zeroth-order theory for most nanoplasmonic structures. It combines Maxwell's equations with a bulk dielectric function $\epsilon(\omega)$, places polarization charge on a mathematically sharp interface, and solves the resulting geometry using Mie theory, transfer matrices, BEM, FEM, or FDTD. For a 2D conductor it uses a local sheet conductivity $\sigma_{2D}(\omega)$; for a TMDC it often uses a local Lorentz-type excitonic sheet susceptibility. These models are efficient and usually predictive when every field-variation scale is much larger than the relevant electronic or excitonic length.

The approximation fails first at the smallest feature—not necessarily at the overall particle size. A 100 nm antenna with a 1 nm gap contains q-components of order $1/g$ and can therefore probe finite screening, spill-out, Landau damping, and tunnelling. Experiments now provide several distinct signatures: EELS blueshifts in small Ag particles [5]; conductance-linked optical crossover in sub-nanometre Au junctions [6, 7]; more than 30% nonclassical shifts in film-coupled resonators [14]; near-field evidence of gap-dependent loss in 2–5 nm MDM waveguides [22]; and broadband Au–air surface-response retrieval [33].

Fully microscopic methods supply the reference physics but not yet a general device solver. Ground-state Hartree–Fock and DFT determine electronic structure; optical response requires TDHF, RPA, TDDFT, GW-BSE, or a transport method such as NEGF. TDDFT can capture the crossover from discrete electron–hole transitions to collective plasmons, atomistic facets, and tunnelling. A 2024 TDDFT+U calculation reproduced measured Ag spectra from 4 to 923 atoms, but the largest particle was only about 3 nm and contained roughly 10,000 active electrons [28]. This illustrates both the power and the multiscale bottleneck.

Feibelman parameters compress the leading microscopic interface response into two complex lengths. Quantum hydrodynamics instead retains continuum electron dynamics through pressure, kinetic-energy, exchange-correlation, and—when solved self-consistently—a diffuse equilibrium density. The attached 2021 paper [17] establishes a particularly useful bridge: a finite surface response already follows from a smooth equilibrium density profile, even if the local constitutive relation is used pointwise. Bulk nonlocal compressibility then adds a separate contribution. This prevents spill-out and nonlocality from being treated as synonyms.

The 2026 anchor paper attacks the remaining computational bottleneck for spherical particles. Its volume-integral, vector-spherical-harmonic formulation reduces SC-HDM to radial blocks and makes the tested self-consistent calculation only about 10% slower than LRA. For a 2 nm sodium sphere it predicts 3.49 eV (LRA), 3.64 eV (hard-wall HDM), and 3.30 eV (SC-HDM), plus a Bennett feature near 4.1 eV [1]. These are numerical predictions, not new experimental measurements.

REVIEW THESIS

The modern field is not choosing between 'classical' and 'quantum' as two monolithic options. It is constructing a hierarchy in which microscopic calculations and measurements provide transferable interface or continuum response functions that can be embedded in full-scale electromagnetic solvers.

1. Introduction: the classical platform

1.1 Local response in metals

In the local-response approximation (LRA), the induced polarization at $\mathbf{r}$ depends only on the electric field at the same point. For a homogeneous isotropic metal,

$$\mathbf{P}(\mathbf{r}, \omega) = \epsilon_0 \chi(\omega) \mathbf{E}(\mathbf{r}, \omega), \quad \mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega).$$

The Drude model supplies the intraband contribution and Lorentz oscillators or measured optical data supply bound-electron and interband contributions:

$$\epsilon_m(\omega) = \epsilon_\infty(\omega) - \omega_p^2 / [\omega(\omega + i\gamma)].$$

This is already a dispersive and lossy material model, but it is local in space: ϵ depends on ω and not on wave vector $\mathbf{k}$. It also assumes a piecewise-constant material profile, so the equilibrium electron density terminates abruptly at the nominal metal surface.

1.2 Nanoparticles, antennas, and thin films

For a small sphere in a dielectric ϵ_d , the quasistatic dipole polarizability is

$$\alpha_0 = 4\pi \epsilon_0 \epsilon_d R^3 (\epsilon_m - \epsilon_d) / (\epsilon_m + 2\epsilon_d),$$

with the familiar dipole condition $\text{Re } \epsilon_m \approx -2\epsilon_d$. Mie theory extends this to arbitrary size and multipole order without abandoning local material response [2]. Nanoantennas are then understood as radiating plasmonic resonators, dimers as hybridized capacitive modes, and thin films or multilayers through Fresnel coefficients and transfer matrices. A planar metal–dielectric interface supports an SPP with local dispersion $k_{\text{SPP}} = k_0 \sqrt{[\epsilon_m \epsilon_d / (\epsilon_m + \epsilon_d)]}$.

The important distinction is that geometric complexity is treated exactly within Maxwell theory while microscopic interface structure is discarded. A highly refined FEM mesh does not repair a local constitutive law; it only solves that local model more accurately.

1.3 Classical sheet models for graphene and TMDCs

An atomically thin material is often represented by a surface current $\mathbf{J}_s = \sigma_{2D}(\omega) \mathbf{E}_{\parallel}$ rather than by an arbitrary thin three-dimensional slab. For doped graphene, a local Kubo or Drude conductivity gives tunable infrared plasmons [3]. For a semiconducting TMDC, a local excitonic sheet model may be written as a sum of Lorentz poles in $\chi_{2D}(\omega)$, with A, B, and higher excitons represented by fitted resonance energies, oscillator strengths, and linewidths.

These are classical electrodynamic embeddings of quantum-derived material parameters. They are not automatically valid at large in-plane momentum. Graphene has a full density response $\chi(\mathbf{q}, \omega)$, and a TMDC exciton has center-of-mass dispersion, nonlocal screening, valley structure, and dark states. The correct high- q object is therefore $\sigma_{2D}(\mathbf{q}, \omega)$ or $\chi_{2D}(\mathbf{q}, \omega)$, not a q -independent sheet constant [4, 40, 27].

1.4 Validity criterion

The controlling comparison is between the shortest electromagnetic variation length L_{field} and each material length ℓ : screening length, Fermi wavelength, spill-out width, mean free path, tunnelling decay length, or exciton coherence radius. Equivalent dimensionless tests are $q\ell$, ℓ/R , ℓ/g , and ℓ/t .

Condition	Interpretation	Expected model status
$q\ell \ll 1$ and $\ell/R, \ell/g, \ell/t \ll 1$	Fields vary slowly across electronic structure	LRA controlled
One ratio ≈ 0.01 – 0.1	Small but measurable interface correction	Perturbative d parameters or HDM
One ratio ≈ 1	Finite screening/spill-out comparable to geometry	SC-HDM/QHT or microscopic benchmark
Wavefunctions overlap across gap	Boundary topology becomes conductive	QCM, TDDFT, or NEGF

The numerical thresholds are indicative, not universal: noble-metal interband screening, facet, dielectric environment, roughness, and frequency can change both magnitude and sign.

2. Scaling down: experimental breakdown of local models

2.1 What breaks, physically

Mechanism	What LRA assumes	Measured or predicted consequence
Finite compressibility	Surface charge can be compressed without spatial cost	Screening layer, hard-wall blueshift, longitudinal modes
Spill-out / spill-in	Equilibrium and induced charge stop at a sharp boundary	Shifted charge centroid, red- or blueshift, effective-gap change
Surface-enabled Landau damping	Loss is fully contained in bulk $\epsilon(\omega)$	q- and size-dependent broadening and propagation loss
Quantum confinement	Electronic spectrum remains bulk-continuous	Discrete transitions, shell/subband effects, non-monotonic size trends
Tunnelling	The dielectric gap is perfectly insulating	Field saturation, mode quenching, charge-transfer plasmons
Atomic chemistry	Surface is chemically inert and geometry-only	Facet, ligand, adsorbate, and bonding-dependent response

These effects are experimentally entangled. A size-dependent linewidth can arise from surface-enabled Landau damping, roughness, grains, adsorbates, radiation, or an incorrect bulk permittivity. A spectral shift can be caused by nonlocal screening, spill-out, a spacer-thickness error, substrate polarization, or particle shape. Strong evidence therefore requires either simultaneous observables or a platform designed to suppress classical uncertainty.

2.2 Experimental evidence: from isolated particles to interfaces

Experiment	Small scale	Observation	What it establishes
Ag-particle EELS (2012) [5]	few-nm particles	Size-dependent plasmon shifts	Bulk local ϵ is insufficient; confinement and surface physics matter
Dynamic Au junction (2012) [6]	sub-nm gap	Optical crossover correlated with conductance	Tunnelling changes the plasmonic boundary condition
Molecular junctions (2014) [7]	molecular bridge	Mode evolution controlled by junction conductance	Conductive coupling, not gap distance alone, controls response
Coaxial apertures (2019) [15]	1–10 nm gaps	Mid-IR shifts incompatible with a purely local model	Nonlocality becomes visible in highly confined extended structures
Film-coupled resonators (2019) [14]	nm-scale Au–AlOx interface	>30% nonclassical shifts; Kreibig broadening fails	Complex interface response can be experimentally retrieved
MDM gap plasmons (2022) [22]	2–5 nm dielectric gaps	s-SNOM complex propagation shows excess gap-dependent loss	Nonlocal damping is enhanced in large-q propagating modes
Single Au nanorods (2023) [26]	single particles	Only modest nonlocal damping relative to coupled gaps	Confinement, not 'nano' size alone, governs visibility
Au–air ellipsometry (2025) [33]	planar interface	Broadband d response, sign changes, Bennett pole near 2.5 eV	Surface response is measurable as a causal complex function

2.3 Breakdown in 2D materials

The failure mechanism in a 2D electron system is analogous but not identical to metal spill-out. Lundeberg et al. used near-field imaging of graphene plasmons in an engineered dielectric–metal environment and obtained a parameter-free match to the full quantum response of the Dirac electron liquid. Short-wavelength response included Fermi-surface deformation plus interaction-dependent inertia and compressibility [4]. This is a direct experimental rejection of a q -independent local sheet conductivity in the accessed regime.

For TMDCs the relevant nonlocality is primarily excitonic rather than hydrodynamic motion of a free-electron gas. Momentum-resolved EELS has now mapped excitonic and exciton-polaritonic features in WSe₂ out to large momentum and shown strong sensitivity to thickness, edges, and patterned geometry [27]. A free-electron HDM should therefore not be transplanted to a semiconducting TMDC; its response must be represented by a momentum-dependent excitonic susceptibility.

DIAGNOSTIC RULE

A discrepancy with LRA is evidence for missing physics, not proof of one unique mechanism. Nonlocality is strongest when both energy and linewidth—or optical and transport/near-field observables—are explained by the same parameter set.

3. Fully microscopic many-body response

3.1 What a quantum calculation must provide

The exact optical response is a two-point susceptibility $\chi(r, r'; \omega)$ obtained from the many-electron system. It contains the spatial extent of screening, discrete electronic levels, interband transitions, electron–hole excitations, surface states, tunnelling, and—in principle—correlations. The computational difficulty is not Maxwell's equations but representing and evolving the electronic state for enough atoms to form the nanostructure.

$$P_i(r, \omega) = \epsilon_0 \int \chi_{ij}(r, r'; \omega) E_j(r', \omega) dr'$$

3.2 Hartree–Fock, TDHF, and RPA

Hartree–Fock (HF) replaces the many-electron wavefunction by one Slater determinant. The Hartree term is a classical mean Coulomb field; the Fock term is nonlocal exchange. Ground-state HF does not itself predict an optical spectrum. Linear optical response requires time-dependent HF (TDHF), configuration-interaction singles, or an equivalent linear-response formulation. Linearized time-dependent Hartree without exchange is the random-phase approximation (RPA), the natural microscopic origin of the electron-gas dielectric response [38].

Unscreened HF exchange is problematic for extended metals and HF omits dynamical correlation. In practical nanoplasmonics, its most important modern role is often through hybrid or range-separated TDDFT functionals that include a fraction of nonlocal Fock exchange. Rabilloud showed that long-range HF exchange changes the assignment of the plasmon-like band in Ag clusters and avoids the excessive d-

electron character produced by semilocal functionals [29]. Thus 'HF captures the quantum response' is too broad; the response theory and screening approximation are decisive.

3.3 DFT and TDDFT

Ground-state density-functional theory (DFT) maps the interacting ground-state density onto Kohn–Sham orbitals in an effective potential. Time-dependent DFT (TDDFT) then obtains the optical response either by real-time propagation after an impulsive field or by solving a linear-response eigenvalue problem. It can include the actual atomic geometry, work function, equilibrium spill-out, d-band screening, ligands, and junction wavefunction overlap.

Its accuracy is functional-dependent. Local or semilocal adiabatic functionals can misplace d bands, underestimate long-range charge transfer, and omit memory-dependent double excitations. DFT+U can correct localized d-shell energies; hybrids add Fock exchange; GW corrects quasiparticle energies; the Bethe–Salpeter equation (BSE) treats electron–hole attraction; NEGF is required when open-boundary transport and optical response must be treated together.

3.4 Theoretical and experimental validation

Study	Quantum content	Validation or lesson
Zuloaga et al. (2009) [30]	Atomistic TDDFT dimer response	Predicts tunnelling-induced mode evolution and charge-transfer plasmons
Weissker & Mottet (2011) [31]	TDDFT noble-metal clusters	Shows size, composition, and d-band effects beyond a bulk ϵ
Scholl et al. (2012) [5]	Single-particle EELS plus quantum/semiclassical comparison	Observed size-dependent Ag shifts that a fixed local ϵ misses
Ma et al. (2015) [32]	Real-time TDDFT of Ag55	Separates collective plasmon motion from single-particle excitations
Zhu et al. (2016) [12]	Experiment with microscopic nanogap theory	Validates quantum corrections in robust sub-nm junctions
Chaudhary & Weissker (2024) [28]	RT-TDDFT+U, 4–923 Ag atoms	Agreement with measured spectra from discrete clusters to a broad 3 nm LSPR

3.5 Why the microscopic route remains expensive

Bottleneck	Scaling consequence	Device-level implication
Atomic representation	Surface atoms, ligands, spacers, and substrates all add variables	A realistic 50–100 nm antenna is beyond routine first-principles response
Occupied/unoccupied states	Response couples many electron–hole pairs; hybrids/GW/BSE add kernels	Broadband spectra and geometry sweeps become expensive
Real-time propagation	Fine time steps and long propagation are needed for energy resolution	Each geometry needs many large Hamiltonian applications
Open boundaries and tunnelling	Electrodes require transport self-energies or large supercells	Joint optical–conductance calculations remain specialized
Material uncertainty	Atomic structure and exchange–correlation choice alter spectra	High cost does not guarantee a transferable material constant

The 2024 Ag result is a useful scale marker: 923 atoms and roughly 10,000 active electrons already represent an unusually large first-principles optical calculation, yet the particle diameter is only about 3 nm. A plasmonic antenna, film-coupled cavity, or metasurface can be tens to hundreds of nanometres across. This gap creates the need for mesoscopic parameterizations.

4. Mesoscopic surface response: Feibelman parameters and spill-out

4.1 Compressing a microscopic interface into two complex lengths

Feibelman's surface-response functions retain the classical bulk permittivities on either side of an interface but replace the zero-thickness induced surface charge and current by their first moments. For a locally planar interface normal to z ,

$$d_{\perp}(\omega) = [\int z \rho_{ind}(z, \omega) dz] / [\int \rho_{ind}(z, \omega) dz],$$

$$d_{\parallel}(\omega) = [\int z \partial_z \rho_{ind}(z, \omega) dz] / [\int \partial_z \rho_{ind}(z, \omega) dz].$$

The real parts mainly encode reactive displacement of the interface response; the imaginary parts encode surface-enabled dissipation. The parameters enter generalized Maxwell boundary conditions or perturbative resonance corrections. To leading order, their influence scales as d/R for curvature and d/g for a gap [35, 13].

4.2 Spill-out, spill-in, and the effective optical boundary

Spill-out means that the equilibrium or induced conduction-electron density extends toward the dielectric beyond the positive ionic background. Spill-in places the screening centroid inside the metal. The sign of a resonance shift is not universal: it depends on both $d_{\perp}$ and $d_{\parallel}$, frequency, interband screening, dielectric environment, reference plane, and mode geometry. For a gap, the electronic response changes the effective optical separation even if the geometric spacer thickness is fixed.

A Bennett mode is a multipole surface excitation supported by a finite-width density profile. It appears as a pole-like surface response and as a sign-changing induced charge across the interface [36]. Its

observation therefore probes the internal structure of the interface rather than only a shifted dipole resonance.

4.3 Attached paper: surface response from equilibrium density profiles

Mortensen, Gonçalves, Shuklin, and co-workers ask whether a dynamical nonlocal calculation is necessary to obtain any finite d parameter. Their answer is no. If the equilibrium density $n_0(z)$ changes smoothly across the interface, then the local Drude permittivity $\epsilon_{\text{LRA}}(z)$ also changes smoothly. Solving local electrodynamics in that inhomogeneous layer produces a finite-width induced charge and therefore finite surface-response moments [17].

THE PAPER'S CONCEPTUAL SEPARATION

Equilibrium inhomogeneity (spill-out or spill-in) can generate a finite surface response within pointwise LRA. Dynamical bulk nonlocality—the finite compressibility described by RPA or HDM—adds another contribution. The two effects can reinforce or cancel and should not be identified with each other.

The paper replaces the classical piecewise-constant approximation $\epsilon_{\text{PCA}}(z)$ by a smooth profile tied to $n_0(z)$. It derives $d_{\perp}$ and $d_{\parallel}$ from integrals of the difference between the smooth and sharp dielectric profiles. For its analytic jellium–vacuum model, the low-frequency response obeys $\text{Re } d_{\perp} \approx z_0$ and $\text{Im } d_{\perp} \approx (\pi/2)a\Omega$, where z_0 sets the density edge, a is the surface-smearing length, and $\Omega = \omega/\omega_p$. The model therefore produces both a centroid shift and surface-assisted damping without a full time-dependent *ab initio* calculation.

Its value is explanatory and computational. It recovers qualitative TDDFT/jellium trends and shows how the sign of the effective response $d_{\text{eff}} = d_{\perp} - d_{\parallel}$ controls the leading $1/R$ shift of a spherical LSP. Its limitation is equally explicit: it omits finite electron-gas compressibility, detailed Friedel oscillations, atomistic facets, and many-body correlations. It is a controlled contribution to d , not a replacement for all microscopic response.

4.4 From theory to measured d parameters

Route	What supplies d	Status and comparison
TDDFT / RPA	Induced charge and current at a planar or atomistic interface	Microscopic reference; expensive and functional/facet dependent
Equilibrium profile [17]	Smooth $n_0(z)$ inserted into local electrodynamics	Analytic, qualitative agreement with <i>ab initio</i> trends; omits dynamic nonlocality
Facet-specific theory [18]	First-principles response for defined noble-metal facets	Shows that d is an interface/facet function, not an elemental constant
Film-cavity retrieval [14]	QNM shifts and linewidths of Au–AlOx resonators	Narrower-band complex interface response; large nonclassical shifts
Broadband ellipsometry [33]	Quantum-corrected Fresnel response of Au–air	Visible–UV d retrieval; spill-in/out sign changes and Bennett pole
Graphene/EELS probes [19, 25]	High- q plasmons or electron-beam spectra	Proposed routes to q - and geometry-sensitive interface metrology

The 2019 and 2025 experiments are especially important because they turn d from a theoretical diagnostic into a measurable complex response. Chen et al. also enforced passivity and Kramers–Kronig consistency, a necessary safeguard for any broadband surface parameter. Agreement with one resonance shift is weaker evidence than a causal complex spectrum that also explains linewidth.

4.5 Domain and failure of the surface expansion

- First-order d -parameter theory requires $|d|$ to remain small relative to curvature radii, gap widths, and modal variation lengths.
- The reference plane must be stated; $d_{\perp}$ and $d_{\parallel}$ can shift under a change of origin although complete observables remain invariant.
- A single $d(\omega)$ may fail at large in-plane q ; dispersive $d(q, \omega)$ or higher surface moments may be required.
- Edges and facet junctions are not locally planar interfaces and may need dedicated edge parameters or microscopic treatment.
- Once electrons tunnel, a purely polarizable interface is incomplete because normal charge transfer changes the boundary topology.

5. Nonlocal continuum response: hydrodynamic models

5.1 Hard-wall hydrodynamic Drude model

The hydrodynamic Drude model supplements the Drude equation with a pressure force. After linearization, one common frequency-domain form is

$$\beta^2 \nabla(\nabla \cdot J) + \omega(\omega + i\gamma)J = i\omega \epsilon_0 \omega p^2 E,$$

where β is of order the Fermi velocity. Transverse fields retain a Drude-like response while longitudinal density waves acquire a finite wave number. An additional boundary condition is required; the standard hard wall imposes $J \cdot n = 0$. This model regularizes charge compression, often blueshifts localized plasmons, and supports confined longitudinal modes. It does not permit equilibrium spill-out or tunnelling.

5.2 Dissipative extensions: GNOR and viscosity

GNOR adds diffusion to the induced-charge dynamics, making the nonlocal coefficient complex and producing size- or q -dependent broadening in addition to a reactive shift [10]. It is computationally efficient and has been used to fit the complex propagation constant of few-nanometre MDM gap plasmons [22]. The fitted diffusion constant is an effective closure: it can impersonate Landau damping or other surface scattering and should not be interpreted as a unique microscopic material constant without independent validation.

5.3 Self-consistent HDM and quantum hydrodynamic theory

Self-consistent hydrodynamics removes the hard density step. It first determines an equilibrium density $n_0(r)$ by minimizing an orbital-free energy functional $G[n]$, then linearizes continuity and force equations

around n_0 . Typical ingredients are the Thomas–Fermi kinetic term, a von Weizsäcker gradient correction, exchange-correlation, electrostatics, and a phenomenological damping term:

$$G[n] = TTF[n] + \lambda TvW[n] + Exc[n] + \text{electrostatic terms}.$$

Because n_0 decays through the interface, the induced charge can spill out and the model can support Bennett-like surface modes. The accuracy is controlled by the kinetic and exchange-correlation functionals, the equilibrium density, and the treatment of the low-density tail [11, 20, 23].

5.4 Theory, use, and experimental validation

Model	Best use	Evidence and limitation
Hard-wall HDM	Fast sensitivity to finite compressibility; longitudinal modes	Robust theory and solver benchmarks; often overpredicts simple blueshifts because spill-out is excluded
GNOR	Nonlocal shift plus effective surface damping in particles and waveguides	Compatible with few-nm gap-plasmon measurements [22]; closure parameter is fitted
SC-HDM / QHT	Diffuse equilibrium density, spill-out/in, surface modes	Benchmarked mainly against TDDFT and known trends [11, 20, 23]; direct material-specific validation remains limited
Surface response	Transfer quantum interface data into BEM/FEM/transfer matrices	Direct retrieval exists for Au interfaces [14, 33]; first-order and q-local limits apply
QCM	Large structures with tunnel-conductive gaps	Calibrated to quantum transport/TDDFT [8]; does not predict chemistry by itself

Hydrodynamic validation is therefore layered rather than absolute. Hard-wall HDM is validated as a controlled long-wavelength electron-gas approximation; SC-HDM is tested against TDDFT charge profiles and size trends; GNOR is compared with measured complex mode propagation; and surface-response parameters are increasingly measured directly. No one continuum closure has yet reproduced noble-metal spill-out, interband screening, Landau damping, atomistic facets, and tunnelling over the full visible–UV range.

5.5 Why QHD is the natural mesoscopic bridge

Quantum hydrodynamics retains only density and current fields, so its degrees of freedom scale with a continuum mesh rather than with occupied and unoccupied electronic states. It can therefore treat structures larger than TDDFT while preserving finite screening and, in self-consistent form, spill-out. It also generates induced charge profiles from which d parameters can be extracted. The central unresolved problem is not the concept but obtaining stable, causal, transferable functionals and low-density boundary treatments for real materials.

6. The 2026 anchor paper: overcoming the SC-HDM bottleneck

6.1 The computational problem

The anchor paper starts from the correct field-level difficulty: SC-HDM is far cheaper than TDDFT in principle, yet a direct three-dimensional differential-equation implementation still introduces extra longitudinal fields, a diffuse exterior density tail, and tightly coupled quantum material operators. This can erase the practical advantage over LRA for repeated spectra or parameter sweeps.

The paper does not report an experiment. It develops and validates a computational formalism for spherical nanoparticles and uses sodium-like material parameters, where a free-electron continuum is most credible.

6.2 Volume-integral and spherical reduction

The scattered field is written through the homogeneous electromagnetic dyadic Green tensor as a volume integral over the polarization $\mathbf{P}$. Substitution into the hydrodynamic constitutive equation yields an integral equation for $\mathbf{P}$ alone. The polarization is expanded in vector spherical harmonics and cubic radial Lagrange functions; Galerkin testing performs the angular integrals analytically.

Design choice	Consequence
Vector spherical harmonics	Exact angular-momentum separation for a sphere
TE/TM/longitudinal basis	TE decouples; TM and longitudinal sectors form small coupled blocks
Cubic radial elements	One-dimensional resolution of diffuse equilibrium and induced density
Frequency-independent material blocks	Quantum operators are assembled once across a spectrum
Volume-integral Green formulation	Radiation is handled without meshing a large exterior domain
Functional solver generator	Different orbital-free $G[n]$ closures can reuse the same algebraic framework

The speedup is therefore symmetry-enabled. It should not be generalized to arbitrary three-dimensional antennas without demonstrating a comparable reduction.

6.3 Validation and physical results

Quantity	Reported result	Interpretation
Validation	LRA and hard-wall HDM match OpenSANS S-matrix spectra within about 0.08% in the reported metrics	The discretization reproduces established spherical solutions
Sphere set	Sodium radii $R = 1\text{--}5$ nm; $\omega_p = 6.04$ eV	Simple-metal test bed for nonlocal and diffuse-surface effects
$R = 2$ nm, LRA	Dipole ≈ 3.49 eV	Sharp-interface local baseline
$R = 2$ nm, HDM	Dipole ≈ 3.64 eV	Hard-wall compressibility pushes screening inward and blueshifts
$R = 2$ nm, SC-HDM	Dipole ≈ 3.30 eV	Diffuse density and induced charge spill outward and redshift
Surface mode	Bennett-type feature near 4.1 eV	Finite-width surface supports sign-changing multipole charge
Runtime	85.69 s LRA; 87.95 s HDM; 94.36 s SC-HDM	About 10% SC-HDM overhead for this spherical implementation

PHYSICAL COMPARISON AT $R = 2$ NM

LRA 3.49 eV $\rightarrow$ hard-wall HDM 3.64 eV $\rightarrow$ SC-HDM 3.30 eV, with a distinct Bennett surface feature near 4.1 eV.

6.4 Extraction of surface response

The paper evaluates a spherical charge-centroid quantity analogous to $d_{\perp}$ and finds a Lorentzian feature at the Bennett resonance. This supports a multiscale route in which SC-HDM generates an interface response that is later used in a cheaper surface-corrected Maxwell solver. The comparison with a planar TDDFT fit is exploratory, however: a finite sphere is not a planar half-space and spatial dispersion $d(q, \omega)$ is omitted.

The SC-HDM response also develops a second zero near 4.62 eV and additional high-energy features whose number depends on the allowed spill-out extent. The authors identify this as a high-energy instability or truncation sensitivity of the selected functional and density-tail treatment. The extracted curve should therefore not yet be used as a universal sodium d parameter.

6.5 Contribution to the field

Established by the paper	Still open
Symmetry-adapted VIE reproduces spherical LRA/HDM benchmarks	Arbitrary 3D shapes, substrates, edges, and touching junctions
SC-HDM can approach LRA runtime for tested spheres	Whether similar efficiency survives without spherical symmetry
Diffuse-interface shifts and a Bennett mode are recovered	Direct experimental validation of the predicted sodium spectra
Finite-sphere surface response can be generated cheaply	Stable, causal, transferable planar $d(q, \omega)$ for real noble-metal interfaces
Framework accommodates alternative energy functionals	Interband realism, tunnelling, and systematic functional calibration

The conceptual arc from the 2021 equilibrium-profile paper to the 2026 VIE paper is therefore coherent. The first identifies how diffuse equilibrium structure contributes to surface response; the second makes a self-consistent continuum calculation of that structure and its dynamics far cheaper in a canonical geometry.

7. Implications for nanostructures and 2D hybrids

7.1 Plasmonic nanoantennas and nanogaps

Feature	Mesoscopic correction	Recommended model
Isolated particle, radius ≥ 10 nm	Often smaller than uncertainty in ϵ , shape, and surface chemistry	Measured LRA baseline; test GNOR/d only if linewidth/shift precision supports it
Tip or neck $\approx 2\text{--}10$ nm	Finite screening and surface loss smooth charge localization	Feibelman response or HDM/GNOR
Capacitive gap $\approx 0.5\text{--}2$ nm	Spill-out changes effective gap; q-dependent response grows	$d(q, \omega)$ plus SC-HDM/QHT benchmark
Conductive sub-nm junction	Tunnelling saturates field and creates charge-transfer modes	QCM calibrated to TDDFT/NEGF or explicit transport
Emitter within $\lesssim 2$ nm of metal	High-k LDOS is suppressed and nonradiative electron–hole loss rises	Surface-response Green tensor [16]

7.2 Thin films and metal–insulator–metal structures

Planar films are strong metrology platforms because thickness and propagation constants can be measured precisely. They also expose parameter degeneracy: roughness, grains, interdiffusion, oxide thickness, and the film's own $\epsilon(\omega)$ can mimic a mesoscopic phase shift. The best experiments therefore use crystalline metals, conformal spacers, and complex observables. Boroviks et al. measured both $\text{Re } q$ and $\text{Im } q$ by s-SNOM; Yang et al. used families of resonators; Chen et al. imposed broadband causality [22, 14, 33].

7.3 Graphene, TMDCs, and hybrid cavities

A metal–2D hybrid contains two different nonlocal subsystems. The metal interface is described by $d_{\perp}/d_{\parallel}$ or QHD; graphene requires $\sigma_{2D}(q, \omega)$; a TMDC requires an excitonic $\chi_{2D}(q, \omega)$, often informed by BSE or a validated polariton model. Combining everything into one local effective permittivity obscures which subsystem supplies the momentum dependence and loss.

In metal–TMDC nanocavities, a mesoscopic metal correction changes cavity frequency, linewidth, mode volume, and emitter overlap, while exciton dispersion and screening change the matter resonance. Room-temperature strong coupling in WSe₂ nanoparticle-on-mirror cavities [41] therefore motivates a two-response calculation: a quantum-informed metal boundary coupled to a nonlocal excitonic sheet.

7.4 Practical model-selection workflow

Step	Question	Action
1	What is the smallest field scale?	Estimate gap g , curvature R , film thickness t , and modal q —not only total device size
2	Is the local baseline constrained?	Use measured ϵ , geometry, roughness, facet, spacer thickness, and uncertainty ensemble
3	Is the missing effect reactive or	Compare both resonance energy and linewidth, or complex

Step	Question	Action
	dissipative?	propagation q
4	Does charge remain confined to each body?	Use d/HDM/QHT if yes; switch to QCM/NEGF/TDDFT if tunnelling occurs
5	Is the interface parameter transferable?	Check reference plane, facet, dielectric, q dependence, curvature, and passivity
6	Is a 2D exciton present?	Use $\chi_{2D}(q, \omega)$ or BSE-informed response separately from metal hydrodynamics
7	What validates the model?	Require more than one observable and compare with a microscopic calculation where tractable

8. Conclusions and current research direction

The local-response approximation remains indispensable, but its success should not be confused with scale invariance. It works because angstrom-scale electronic structure is usually invisible to the field. Extreme gaps, thin films, large- q plasmons, and atomically thin materials deliberately remove that separation of scales.

Microscopic DFT/TDDFT and HF-derived response theories establish what the electron system actually does, but they are too expensive for most full photonic devices. Feibelman parameters and QHD are complementary reductions of the same two-point response: the former integrates the interface into surface moments; the latter retains a continuum density/current field. The attached 2021 paper shows that equilibrium spill-out contributes to the surface response even without dynamic nonlocality. The 2026 paper then demonstrates that a self-consistent continuum treatment can be almost as cheap as LRA in spherical symmetry.

The near-term research program is therefore a calibrated multiscale pipeline: obtain n_0 , χ , or d from TDDFT, QHD, or measurement for a defined interface; enforce causality and passivity; retain q dependence when required; and deploy that response in BEM/FEM models of realistic antennas, films, and 2D hybrids. The hardest unresolved boundary is the conductive angstrom junction, where surface polarization, chemistry, and quantum transport cease to be separable.

Annotated primary literature

The references below prioritize original theoretical and experimental papers used to construct the model hierarchy. Review articles and textbooks are intentionally not the main evidentiary basis.

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Appendix A. Compact terminology

Term	Meaning in this review
LRA	Local response: ϵ depends on frequency but not wave vector; sharp interfaces.
PCA	Piecewise-constant approximation for the material profile at an interface.
RPA	Random-phase approximation; linearized time-dependent Hartree screening.
TDHF	Time-dependent Hartree–Fock; linear response includes Fock exchange but not full correlation.
TDDFT	Time-dependent density-functional theory for frequency- or time-domain electronic response.
HDM	Hard-wall hydrodynamic Drude model with finite compressibility and longitudinal waves.
GNOR	Hydrodynamic response with diffusion-like nonlocal damping.
SC-HDM / QHT	Self-consistent hydrodynamic continuum with equilibrium density and energy functional.
$d_{\perp}, d_{\parallel}$	Complex Feibelman surface-response centroids for normal charge and tangential current.
Spill-out / spill-in	Displacement of equilibrium or induced electronic density relative to the ionic boundary.
Bennett mode	Multipole excitation of a finite-width electronic surface layer.
QCM	Quantum-corrected model using an effective conductive junction.
$\chi^2_{2D}(q, \omega)$	Momentum- and frequency-dependent sheet susceptibility of a 2D material.

Appendix B. Anchor-paper reproducibility facts

- Material: sodium-like Drude metal with reported plasma frequency 6.04 eV.
- Geometries: homogeneous spheres with radii 1, 2, 3, 4, and 5 nm.
- SC-HDM functional: Thomas–Fermi + full von Weizsäcker ($\eta = 1$) + Perdew–Zunger LDA exchange–correlation.
- Equilibrium density: prescribed sigmoid profile with a truncated spill-out allowance s .
- Basis: vector spherical harmonics angularly and cubic Lagrange functions radially; Galerkin testing.
- Validation: OpenSANS spherical S-matrix spectra for LRA and hard-wall HDM.
- Key 2 nm energies: 3.49 eV LRA, 3.64 eV HDM, 3.30 eV SC-HDM, and a Bennett feature near 4.1 eV.
- Runtime means over 20 runs: 85.69 s, 87.95 s, and 94.36 s, respectively.
- Interpretation limit: finite-sphere centroid versus planar TDDFT fit; $d(q, \omega)$ omitted; high-energy response depends on spill-out truncation.

End of review